

Results Guide — ACh imaging in striatum (PG_005)

What the data show for each claim, then where to find it.

Results copied from the OIST clusters on 2026-09-27 into [results/](#) (spike-aligned run with `BASELINE_SIGMA_MULT = 2.0`; a 1.5 re-run is in progress).

Terms in *italics* are explained in [Key terms](#).

1. Claims

Claim 1 — Spontaneous ACh forms local zones that cover the striatum, but not in phase

What we see:

- 70 / 105 recordings (10X) have spontaneous *zones* — 1,330 zones in total.
- Median per recording: 15 zones, zone area $\sim 102,700 \mu\text{m}^2$, one event every $\sim 9 \text{ s}$ (0.11 Hz).
- Zones cover a median 56 % of the striatum (*coverage* 0.56).
- Good example: [2025_06_11-0003](#) — 26 zones, coverage 0.64.
- 35 recordings have no zones at all. They are whole days: 2025_12_14 (16), 2025_11_27 (10), 2025_12_18 (7), 2024_10_11 (2, iAChSnFR).

Where to look:

- [spontaneous/spontaneous_summary.xlsx](#)
 - sheet `recordings` → `n_zones`, `median_area_um2`, `median_freq_hz`, `striatum_coverage`
 - sheet `zones` → every zone: area, number of events, frequency
- [spontaneous/{rec}_ZONE_MAPS.tif](#) (multi-page)
 - page 1: max projection + all zones in colour, numbered; dashed line = striatum boundary; axes = ventral–dorsal, medial–lateral
 - page 2 onward: one page per detection frame; title = frame / time, *threshold*, max z, active zone; blue = zone outline, white = the hotspot in that frame
- [spontaneous/{rec}_ZONES.xlsx](#) → sheets `trace_corr_groups` / `proximity_groups` / `isolated_tracks` list the **active frames** of each zone

Still open:

- **"Not in phase" is not measured yet.** The active frames of each zone are available; a timing comparison between zones is still to do.
 - [2025_11_27-0033](#): one very large hotspot in mid-recording (frame 1078) was removed by the 80 % filter — worth a look.
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Claim 2 — One spike of one neuron releases a hotspot $\geq$ natural spontaneous hotspots

What we see:

Spike-triggered hotspots (*MED*), detected recordings only:

OBJ	Detected / total	Reliability (median)	Spike-frame hotspot (median)	Lasting time (median)
10X	41 / 92	52 %	~72,600 μm^2	73 ms
40X	9 / 9	8 %	~34,200 μm^2	60 ms
60X	17 / 62	13 %	~21,000 μm^2	74 ms

MED hotspot vs. spontaneous zones of the same field of view (10X, [area_comparison.xlsx](#)):

Group	n	Median	Q1 – Q3
A — <i>estim_induced</i> MED	26	74,364 μm^2	13,412 – 172,568
B — <i>spontaneous</i> MED	15	145,385 μm^2	93,302 – 177,437
C — spontaneous zones (not at the MED spot)	776	45,228 μm^2	30,864 – 73,999

Test (Mann-Whitney U, two-sided)	p
A vs C	0.79
B vs C	0.0001
A vs B	0.11

- Per recording, the MED is larger than a median 88 % of that recording's zones; in 27 / 40 recordings it is larger than at least half of them.
- Group A is split: many *estim* MEDs are ~200,000 μm^2 (e.g. the 2025_12_15 and 2025_06_11 cells), others are ~10,000 μm^2 (e.g. 2025_11_08, 2025_11_13, 2026_01_08).
- Good example: [2025_12_15-0012](#) (*estim_induced*) — 8 / 8 segments detected, 204,107 μm^2 on the spike frame, τ = 83 ms.
- Skipped recordings: 96 × "no significant ACh detection", 38 × "no valid segments" (the neuron fired too densely for a clean baseline).

Where to look:

- [area_comparison.xlsx](#)
 - MED → one row per recording: MED area, matched zone, [pct_zones_smaller](#) (% of that recording's zones smaller than the MED)
 - Zones / Zone events → zone sizes (per zone / per event)
 - Groups, Group tests → the two tables above
- [ana_20260922_000_deigo_cells.xlsx](#)
 - Summary → neurons with a detected hotspot (19 / 36)
 - Spatial, Temporal → hotspot size and lasting time across neurons
 - Neurons → which recording of which neuron was detected
 - Skipped → why a recording has no result
- [spatial/{rec}_SPATIAL.png](#)
 - top: hotspot area per frame (star = *critical frame*, green dashed = decay fit with τ)
 - middle: CAT masks, frame -4 to +4 (pink = hotspot counted, × = centroid)
 - bottom: all spike waveforms overlaid

- [reliability/{rec}_RELIABILITY.png](#) → one tile per *segment*; green frame = hotspot found; title = reliability
- [reliability/{rec}_VM_SUCCESS_FAIL.png](#) → Vm of successful vs. failed segments; black dots = AP threshold
- [spikes/ABF_*_spike_analysis.png](#) → whole Vm trace; green dots = spikes used
- [median/*_MED.tif](#), [categorized/*_CAT.tif](#) → the movies behind the figures (open in ImageJ)

Still open:

- Why group A splits into large and small hotspots (cell? slice? stimulation?).
- The per-objective table above was computed from [results.db](#); it is not in a sheet yet.

Claim 3 — Released ACh stays local instead of spreading as a wave

What we see:

Flow pattern per frame pair, in the spike → spike+4 window:

OBJ	Recordings	Anisotropic	Source / sink
10X	41	61 %	39 %
40X	9	81 %	19 %
60X	17	80 %	20 %

- Example [2025_12_15-0012](#): source → sink → sink → sink → anisotropic L 25° D. The hotspot appears, shrinks back in place and fades; it does not travel across the field.

Where to look:

- [ana_20260922_000_deigo_cells.xlsx](#), sheet [Flow pattern](#) → the table above
- [results.db](#), table [flow_pairs](#) → one row per frame pair: pattern, drift / spread (μm per frame), DV / ML direction
- [flow/{rec}_FLOW.png](#)
 - row 1: flow arrows on the MED (gray = z), inside the striatum
 - row 2: flow arrows only inside the CAT hotspot
 - row 3: flow speed (μm/s)
- [flow/{rec}_STREAMLINES.png](#)
 - row 1: streamlines on the MED
 - row 2: streamlines inside the hotspot; title ends with the pattern; last panel has a D–V–M–L crosshair

Still open:

- No single "stays local" number yet (e.g. how far the hotspot centre moves vs. its size).

2. Key terms

Term	Meaning
Hotspot	a connected patch of pixels above threshold in one frame
Threshold	from the intensity histogram: $\text{thr} = \text{peak} + 2 \times \sigma$ (figure titles: $\text{thr} = c + 2.0 \times \sigma = \text{value}$)
z (gray colour bar)	intensity in units of baseline σ ; 1 = noise level, > 5 = strong signal
Segment	a short clip around one spike (10 frames before, 10 after)
MED	pixel-wise median of all segments of one recording — the "typical" response to one spike
CAT	the MED turned into a binary hotspot mask (window 201 px, density ≥ 0.1 , threshold 2σ , objects $\geq 2,700 \mu\text{m}^2$)
Reliability	% of single segments that show a hotspot on their own
Critical frame	the spike frame or the spike+1 frame, whichever has the larger hotspot
Lasting time	decay time constant τ of the hotspot area after the peak (ms)
estim_induced / spontaneous	origin of the spike: evoked by a current pulse (> 200 pA in ABF CH2) vs. fired on its own
Zone	a place that lights up repeatedly; the union of all hotspot footprints that belong together
trace-corr / proximity / isolated	how hotspots were grouped into a zone: correlated time traces ($r \geq 0.95$) / close by / alone
Coverage	zone area inside the striatum $\div$ striatum area
Optical flow (TV-L1)	for every pixel, how far the image content moved between two frames (px per frame)
Flow pattern	fit of the flow inside the CAT hotspot: source (spreads out), sink (shrinks in), anisotropic (drifts one way, e.g. L 25° D = lateral, tilted 25° toward dorsal)

3. How the results were produced

Spontaneous ([spontaneous_analysis.py](#), saion job 4734460, 105 × 10X):

1. Threshold each frame ($\text{peak} + 2\sigma$); keep hotspots between 1 % and 80 % of the frame.
2. Chain hotspots across frames into tracks (centroids < 115 px apart).
3. Group tracks into zones (trace correlation, then proximity).
4. Per zone: area, number of events, frequency. Per recording: coverage of the striatum.

Spike-aligned ([ach_domain_analysis.py](#), deigo job 45233657, 163 recordings):

1. Detect spikes in the patch-clamp Vm (ABF); keep spikes with enough quiet time around them.
2. Cut a ± 10 -frame segment around each spike → reliability (single segments) and MED (median).
3. MED → CAT hotspot mask → hotspot size on the spike / spike+1 frame, lasting time.

4. TV-L1 optical flow between consecutive MED frames → flow pattern per frame pair.

Area comparison (`area_comparison.py`, run locally on the copied `results.db`):

- 1. MED hotspot size = the larger of the spike / spike+1 frame.
- 2. Zone size = median per-event area of each spontaneous zone.
- 3. The zone nearest the MED hotspot (< 50 px) is set aside; the others form group C.

4. Quick facts

Item	Value
Imaging	1200 frames at 20 Hz (1 frame = 50 ms), 1024 × 1024 px
Sensors	GACH3.0 (most), iAChSnFR (2024_10_11)
Objectives	10X (≈ 1.33 μm/px), 40X, 60X
File name	<code>2025_12_15-0012_A1S3RC1_BIEXP_ALS_...</code> = date - image serial _ animal 1, slice 3, right hemisphere, cell 1 _ preprocessing
<code>BIEXP_ALS</code>	movie preprocessed with a bi-exponential detrend (bleaching) + ALS baseline removal (slow drift)
Striatum boundary	hand-drawn outlines in <code>data/bd_20260922_000.json</code> (used for coverage and DV / ML directions)
Logs	<code>results/logs/</code> — both <code>.err</code> files empty, no crashes

5. Folder map

results/		
└─ results.db		spike-aligned numbers, one row per recording (SQLite)
└─ ana_20260922_000_deigo_cells.xlsx		spike-aligned summary sheets
└─ area_comparison.xlsx		MED hotspot vs. spontaneous zones (claim 2)
└─ spikes/	ABF_*_spike_analysis.png	Vm trace + which spikes were used
└─ reliability/	*_RELIABILITY.png, *_VM_SUCCESS_FAIL.png	
└─ median/	*_MED.tif	spike-aligned median movie (21 frames)
└─ categorized/	*_CAT.tif	binary hotspot movie of the MED
└─ spatial/	*_SPATIAL.png	hotspot size over time
└─ flow/	*_FLOW.png, *_STREAMLINES.png	how the hotspot moves / spreads
└─ spontaneous/		
└─ spontaneous_summary.xlsx		one row per recording + one row per zone
└─ *_ZONES.xlsx		per-recording zone details

— *_ZONE_MAPS.tif	multi-page zone figure
— footprints/ *_ZONES.npz	zone masks (for re-analysis)
— mask/ *_ZONE_MASK.tif	thresholded movie (for re-analysis)

6. Known issues

- 8 recordings have `has_region = 1` in `results.db` but 0 % reliability and no MED / CAT files. Treat them as *not detected* (`area_comparison.py` already skips them):
 - 10X: 2025_12_14-0020, -0022, -0023, 2025_12_18-0023
 - 60X: 2025_04_03-0035, -0036, -0037, 2025_11_08-0027
- 4 neurons never appear in `Neurons` (so 36, not 40, in `Summary`): all of their recordings were "no valid segments".